

HyeongYeop Kang

CONTACT INFORMATION	IIIXR Laboratory, Department of Computer Science and Engineering Rm 502, Korea University 145 Anam-ro, Seongbuk-gu Seoul, 02841 Republic of Korea +82-2-3290-4608 siamiz_hkang@korea.ac.kr
RESEARCH INTERESTS	Extended Reality (VR/AR/MR), Embodied AI & Agents, Computer Graphics, World Models, Multimodal AI, Human-Computer Interaction
EDUCATION	Korea University , Seoul, Korea Ph.D., Computer Science and Engineering, Mar. 2012 - Aug. 2017 • Thesis: <i>Multi-resolution Terrain Rendering with Unlimited Detail and Resolution</i> • Advisors: JungHyun Han, Ph.D B.S., Computer and Communication Engineering, Mar. 2008 - Feb. 2012
POSITIONS HELD	Associate Professor Mar. 2025 - present Department of Computer Science and Engineering, Korea University Assistant Professor Sep. 2024 - Feb. 2025 Department of Computer Science and Engineering, Korea University Assistant Professor Mar. 2020 - Aug. 2024 Department of Software Convergence, Kyung Hee University <i>Chair, Graduate Department of Software Convergence</i> (Mar. 2021 - Aug. 2024) Assistant Professor Sep. 2019 - Feb. 2020 Software, Media and Industrial Engineering Department, Kangwon National University Research Professor Sep. 2018 - Aug. 2019 Next Generation Game Research Center, Korea University Research Professor Sep. 2017 - Aug. 2018 Computer Science and Engineering Research Center, Korea University
REFEREED CONFERENCE PUBLICATIONS	18. SeungJeh Chung, GeonHo Park, Misong Kim, and HyeongYeop Kang* , CAdam: Context-Adaptive Moment Estimation for 3D Gaussian Densification in Generative Distillation, ACM SIGGRAPH , Jul. 2026. 17. SeongRae Noh, SeungWon Seo, Gyeong-Moon Park*, and HyeongYeop Kang* , Edit-As-Act: Goal-Regressive Planning for Open-Vocabulary 3D Indoor Scene Editing, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) , Jun. 2026. 16. SeungWon Seo ¹ , SooBin Lim ¹ , SeongRae Noh ¹ , Haneul Kim, and HyeongYeop Kang* , From Assumptions to Actions: Turning LLM Reasoning into Uncertainty-Aware Planning for Embodied Agents, International Conference on Learning Representations (ICLR) , Apr. 2026. 15. Seong-Eun Hong ¹ , Jaeyoung Seon ¹ , Juyeong Hwang, JongHwan Shin, and HyeongYeop Kang* , Event-T2M: Event-level Conditioning for Complex Text-to-Motion Synthesis, International Conference on Learning Representations (ICLR) , Apr. 2026.

14. Seong-Eun Hong, Juyeong Hwang, Ryunha Lee, and **HyeongYeop Kang***, ORACLE: Orchestrate NPC Daily Activities using Contrastive Learning with Transformer-CVAE, **Computational Visual Media Conference (CVM)**, Apr. 2026.
13. Seong-Eun Hong, Soobin Lim, Juyeong Hwang, Minwook Chang, and **HyeongYeop Kang***, BiPO: Bidirectional Partial Occlusion Network for Text-to-Motion Synthesis, **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, Mar. 2026 (Oral Presentation, Top 4%).
12. Juyeong Hwang, Seong-Eun Hong, Jaeyoung Seon, and **HyeongYeop Kang***, How Does a Virtual Agent Decide Where to Look? Symbolic Cognitive Reasoning for Embodied Head Rotation, **ACM SIGGRAPH Asia**, Dec. 2025.
11. DongHeun Han, Byungmin Kim, RoUn Lee, KyeongMin Kim, Hyoseok Hwang, and **HyeongYeop Kang***, ForceGrip: Reference-Free Curriculum Learning for Realistic Grip Force Control in VR Hand Manipulation, **ACM SIGGRAPH**, Aug. 2025.
10. ByungMin Kim, DongHeun Han, and **HyeongYeop Kang***, Shaping the Future of VR Hand Interactions: Lessons Learned from Modern Methods, **IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)**, Mar. 2025.
9. Seungwon Seo¹, Seongrae Noh¹, Junhyeok Lee, Soobin Lim, Won Hee Lee, and **HyeongYeop Kang***, REVECA: Adaptive Planning and Trajectory-based Validation in Cooperative Language Agents using Information Relevance and Relative Proximity, **Association for the Advancement of Artificial Intelligence (AAAI)**, Mar. 2025 (Oral Presentation, Top 5%).
8. Seungjeh Chung, Joohyun Park, and **HyeongYeop Kang***, 3DStyleGLIP: Part-Tailored Text-Guided 3D Neural Stylization, **Pacific Graphics (PG)**, Oct. 2024.
7. Joohyun Park, Yujin Jeon, HuiYong Kim, SeungHwan Baek, and **HyeongYeop Kang***, P-Hologen: An End-to-End Generative Framework for Phase-Only Holograms, **Pacific Graphics (PG)**, Oct. 2024.
6. SooBin Lim, SeungWon Seo and **HyeongYeop Kang***, DARAM: Dynamic Avatar-Human Motion Remapping Technique for Realistic Virtual Stair Ascending Motions, **ACM SIGGRAPH**, Jul. 2023.
5. Geonsun Lee, **HyeongYeop Kang**, Jongmin Lee and Junghyun Han*, “A User Study on View-sharing Techniques for One-to-Many Mixed Reality Collaborations,” **IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)**, Mar. 2020, Atlanta, Georgia, United States (**Best conference paper nominee**).
4. **HyeongYeop Kang**, Geonsun Lee and Junghyun Han*, “Obstacle detection and alert system for smartphone AR users,” **ACM Symposium on Virtual Reality Software and Technology (ACM VRST)**, Oct. 2019, Sydney, Australia.
3. **HyeongYeop Kang**, Geonsun Lee and Junghyun Han*, “Visual Manipulation for Underwater Drag Force Perception in Immersive Virtual Environments,” **IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)**, Mar. 2019, Osaka, Japan (**Best conference paper nominee**).
2. **HyeongYeop Kang**, Geonsun Lee, Dae Seok Kang, Ohung Kwon, Jun Yeup Cho, Ho-Jung Choi and Junghyun Han*, “Jumping Further: Forward Jumps in a Gravity-reduced Immersive Virtual Environment,” **IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)**, Mar. 2019, Osaka, Japan (**Best conference paper nominee**).

REFEREED
JOURNAL
PUBLICATIONS

1. **HyeongYeop Kang**, Geonsun Lee, Seongsu Kwon, Ohung Kwon, Seongpil Kim and JungHyun Han*, “Flotation Simulation in a Cable-driven Virtual Environment - A Study with Parasailing,” **ACM Conference on Human Factors in Computing Systems (CHI)**, Apr. 2018, Montreal, Canada.
13. Jongwook Jeong, Myeongseok Kwak, and **HyeongYeop Kang***, Visual Interfaces to Mitigate Eye Problems in a Virtual Environment via Triggering Eye Blinking and Movement, **IEEE Transactions on Human-Machine System (THMS)**, Mar. 2025.
12. KyeongMin Kim and **HyeongYeop Kang***, DAMO: A Deep Solver for Arbitrary Marker Configuration in Optical Motion Capture, **ACM Transactions on Graphics (ToG)** (Siggraph 2025 invited), Sep. 2024.
11. SeungJeh Chung, Taehun Lee, Bora Jeong, Jongwook Jeong and **HyeongYeop Kang***, VRCAT: VR Collision Alarming Technique for User Safety, **The Visual Computer**, Vol. 39, No. 7, Jul. 2023, pp. 3145-3159.
10. DongHeun Han, Roun Lee, KyeongMin Kim and **HyeongYeop Kang***, VR-HandNet: A Visually and Physically Plausible Hand Manipulation System in Virtual Reality, **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, Mar. 2023.
9. MinYeong Seo and **HyeongYeop Kang***, VR Blowing: A Novel Interaction Method for Blowing Air in the Virtual Reality, **IEEE Transactions on Visualization and Computer Graphics (TVCG)**, Jan. 2023.
8. Cheolwoo Lee, Seokhee Jeon, Waseem Hassan and **HyeongYeop Kang***, VR Unseen Gaze: Inducing Feeling of Being Stared At in Virtual Reality, **Virtual Reality**, Jan. 2023.
7. MinYeong Seo and **HyeongYeop Kang***, Towards virtual stair walking, **The Visual Computer**, Vol. 37, No. 9, Jun. 2021.
6. DongHeun Han, ChulWoo Lee and **HyeongYeop Kang***, Gravity Control-based Data Augmentation Technique for Improving VR User Activity Recognition, **Symmetry**, Vol. 13, No. 5, May 2021.
5. Jong-chul Yoon and **HyeongYeop Kang***, Interactive learning in the classroom: A mobile augmented reality assistance application for learning, **Computer Animation and Virtual Worlds**, Jan. 2021.
4. **HyeongYeop Kang** and Junghyun Han*, SafeXR: Alerting Walking Persons to Obstacles in Mobile XR Environments, **The Visual Computer**, Vol. 36, No. 10-12, pp. 2065-2077, Jul. 2020.
3. **HyeongYeop Kang**, Yeram Sim and Junghyun Han*, “Terrain Rendering with Unlimited Detail and Resolution,” **Graphical Models**, Vol. 97, May 2018, pp. 64-79.
2. **HyeongYeop Kang** and Junghyun Han*, “Feature-preserving procedural texture,” **The Visual Computer**, Vol. 33, No. 6-8, Jun. 2017.
1. **HyeongYeop Kang**, Hanyoung Jang, Chang-Sik Cho and JungHyun Han*, “Multi-resolution terrain rendering with GPU tessellation,” **The Visual Computer**, Vol. 31, No. 4, Apr. 2015, pp. 455-469.

BOOK	The Future of the Metaverse: Hyper-Realistic Technologies Tak Woo, Seokhee Jeon and HyeongYeop Kang , Kyung Hee University Communication & Press, ISBN = 9788982227226	2022
GOVERNMENT GRANTS	Korea Creative Content Agency (KOCCA) Development of AI-based Virtual Artist Performance Production Technology for Interactive Audience Engagement, Co-investigator	2026 - 2029
	National Research Foundation of Korea (NRF) MetaSelf-RL: Metacognitive Self-Evolving Framework for High Dimensional, Multi Stage, and Unstructured Physical Interaction Tasks, Principal Investigator	2025 - 2028
	National Research Foundation of Korea (NRF) Gaussian VR Space Representation for Realistic Rendering and Interaction in Dynamic Lighting, Principal Investigator	2025 - 2029
	National Research Foundation of Korea (NRF) Obstacle Detection and Alert System for Multi-user Extended Reality, Principal Investigator	2020 - 2023
	Korea Creative Content Agency (KOCCA) Telepresence and Remote Interaction, Participant (as a research professor)	2017 - 2020
	National Research Foundation of Korea (NRF) Drone Navigation Assistance System Using 360 Camera and HMD, Participant (as a research professor)	2016 - 2020
INDUSTRY GRANTS	Pearl Abyss AI-Driven Rendering Pipeline for Next-Generation Game Production, Principal investigator	2025 - 2026
	FunctionBay Battlefield Simulation Software Development, Principal investigator	2025 - 2026
	LG Electronics Understanding Multimodal PDF Documents Using LLMs, Principal investigator	2025 - 2026
	FunctionBay Manned and Unmanned Integrated Combat System Simulation Development, Co-investigator	2023 - 2024
	Korea Electronics Technology Institute Development of Motion Generation Technology for Anti-Intrusion Avatar in Choreography Animation, Principal Investigator	2023
	Korea Electronics Technology Institute	2023

An Advanced Module for Generating IMU and Video-Based Training Data,
Principal Investigator

NCSOFT 2023 - 2024

Persona-aware NPC gesture motion synthesis,
Principal Investigator

NCSOFT 2022 - 2023

Style-controllable Gesture Synthesis Using Persona Categorization,
Principal Investigator

Samsung Future Technology Promotion Project 2022 - 2025

Phase Hologram Generation and Compression based on Circular Representation and
Randomness Regularization,
Co-investigator

Pearl Abyss 2021 - 2023

Pearl Abyss × KHUSWCON Fellowship Project,
Principal Investigator

PATENTS

10. Method and Apparatus for Torque Control of Virtual Reality Hand Model Based on Curriculum Learning with Progressive Difficulty Control
(KR 10-2025-0145356, 2025)
9. Method and Apparatus for Agent Control through Cooperative Plan Validation Based on Information Relevance and Relative Proximity Using LLM in Multi-Agent Cooperative Environment Based on Virtual Reality
(KR 10-2025-0137820, 2025)
8. Virtual Agent's Daily Activity Plan Generation System and Method of Operation Thereof
(KR 10-2024-0054714, 2024)
7. Device and Method for Generating Generative Model-Based Phase Holograms
(KR 10-2024-0019602, 2024)
6. A Method of Reconstructing Human Pose from Unordered Sparse Point Cloud and a Computer Program for Executing the Same
(KR 10-2023-0157467, 2023)
5. Avatar-Human Motion Remapping Method and a Computer Program for Executing the Same
(KR 10-2023-0152298, 2023)
4. Apparatus for Virtual Hand Motion Generation based on Controller and Method for Thereof
(KR RS-2022-00155911, 2022)
3. Virtual Reality Service Providing User Device, Method and System
(KR 1-1-2022-0479539-49, 2022)
2. Method for Rendering Terrain
(US 9959670 B2, 2018)
1. Method for Rendering Terrain
(KR 002497, 2014)

AWARDS & HONORS	Top Reviewer Recognition (ICML 2026) Awarded <i>Gold Reviewer</i>	May. 2026
	OSSTEM Technical Paper Award (OSSTEM IMPLANT)	Jul. 2023
	Outstanding Presentation Award (ACM SIGCHI Korea Chapter)	Apr. 2018
	Virtual Training System Award (Ministry of Trade, Industry and Energy)	
	• Virtual Training for Reduced Gravity Environment	Dec. 2016
	Qualcomm Innovation Award (Qualcomm)	
	• Terrain Searching with Fully Convolutional Neural Network	May. 2016
	Global Ph.D. Fellowship (National Research Foundation of Korea)	2012 - 2013
	Nexon Open Studio, 1st place (Nexon)	
	• Virtual Economy: Correlation of Estate, Stock and Labor Market	Oct. 2010
	Game Developing Competition, 1st place (Nexon)	
	• The Game of Risk: Game Theories in Online Game Economy	May 2010
	Game Developing Competition, 1st place (Nexon)	
	• Mazer: Creating a Maze	May 2009
	National Science & Tech. Scholarship (Korea Student Aid Foundation)	2008 - 2011
ADVISING & MENTORING	Geonsun Lee	Sep. 2017 - Aug. 2020
	University of Maryland, Ph.D. student	
	Yeram Sim	Mar. 2014 - Aug. 2016
	Nexon Korea, Game programmer	
TEACHING	Department of Computer Science and Engineering, Korea University	2025 - present
	Computer Graphics	
	Visualization	
	Department of Software Convergence, Kyung Hee University	2020 - 2024
	Game Engine Basics (Unity, Unreal)	
	Game Engineering	
	Game Player Experience Design	
	Game Graphics Programming (OpenGL, DirectX, Vulkan, CUDA)	
	Design Thinking	
	Game Engineering	
	Software, Media and Industrial Engineering, Kangwon National University	2019 - 2020
	Basics of C++ Language	
	Digital Logic Design	
PROFESSIONAL SERVICE	Conference Leadership & Committees	
	IEEE VR 2026, Exhibition & Sponsor Chair	
	SIGGRAPH Asia 2025, Technical Communications & Posters Committee	
	SIGGRAPH Asia 2025, XR Program PC Member	
	SIGGRAPH 2025 Chapter in Seoul, Steering Committee	
	IEEE VR 2025, Demo Awards	
	KCGS 2025, Organizing Committee	
	KCGS 2024, Organizing Committee	
	KCGS 2023, Operational Committee	
TALKS	AsiaGraphics Workshop (SIGGRAPH Asia 2025)	Dec. 2025
	Seoul Urban AI Forum (Panel Talk)	Oct. 2025
	Korea Electronics Technology Institute (KETI)	Jun. 2025

LG Electronics

Apr. 2025

French Institute for Research in Computer Science and Automation (INRIA) Mar. 2025